Storage Lab: Multi-Tenant Backup Job Scheduling & Lock Conflict Resolution

Duration : 30 Min.

Scenario

Two automated backup scripts (a database snapshot script and a file-level archive script) run simultaneously at 01:00 AM, both attempting to lock the shared underlying storage volume /dev/vg_app/lv_data. The database snapshot fails with 'ERROR: Volume lock already acquired by another process'. The engineer must analyze the lock conflict, reschedule the jobs into sequential, non-overlapping windows with sufficient buffer, and implement lock-file verification.

Task Objectives

Perform the following backup schedule optimization and concurrency control tasks:

> **Pre-Staged Files in `student_workspace/`:** - **`backup_durations.log`**: Historical backup execution duration logs. - **`safe_backup.sh`**: Starter backup wrapper script.

1. Review `backup_durations.log` to identify schedule overlap and lock collisions between concurrent backup jobs.
2. Re-architect the backup cron schedules into sequential, non-overlapping windows with adequate safety buffers.
3. Implement an advisory file locking mechanism in `safe_backup.sh` to prevent concurrent job collisions.
4. Verify that concurrent execution attempts handle lock contention gracefully.

Instructions to Perform the Task

1. Open your assessment **Terminal** or storage management CLI interface.
2. Execute the required storage management commands, volume operations, or configuration edits.
3. If configuration files, backup manifests, or diagnostic reports are required, save them inside the `student_workspace` directory.
4. Verify your storage configurations locally by executing: `bash python run.py`
5. Once verified, return to the platform dashboard and click **"Run Tests"** / **"Full Eval"** to evaluate your score!

Validation

Once your storage tasks are executed, return to the platform dashboard and click the **"Run Tests"** button. This will automatically evaluate your changes and generate your score!

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Schedule conflict detection: accurately identify the 01:00 AM race condition between snapshot and archive jobs	3 Marks
TC2	Window optimization: cron schedule adjusted to separate jobs based on observed completion times	2 Marks
TC3	Advisory lock implementation: flock wrapper implemented to prevent concurrent execution collisions	3 Marks
TC4	Verification test: execute safe_backup.sh concurrently in two subshells; verify second instance exits cleanly with warning	2 Marks

Total Score: 10 Marks

Important Notes

- This is an **Auto-Evaluated** storage assessment.
- Ensure all created volumes, files, and scripts are properly configured and saved before submitting.